
ZEN-i¹

Team Name: REMMZ		
Moegamat Rashaad Samsodien	Entisaar Elfadl	Misha Dick
Mikaeel Naidoo	Zahra Salie	

Abstract

Traditional credit-scoring models rely heavily on rigid banking histories, leaving individuals with non-traditional incomes financially excluded. In South Africa, a staggering 16 million citizens remain "credit invisible," unable to access credit without a prior history. When these individuals turn to micro-lenders, late payments trigger predatory flat fees that worsen systemic debt. This paper introduces Zen-i, an inclusive Buy Now, Pay Later (BNPL) platform engineered for flexible-income earners. Built using Node.js, React, Open Payments, and Interledger Wallets, Zen-i implements a "Fintech Safety Net" that replaces destructive flat fees with an interest-bearing third-party repayment model. It also deploys a Credit Score Simulator that generates a cash-flow-based "Confidence Score" to mirror bureau algorithms. Our empirical data shows that while first-time borrowers exhibit high initial default rates (21.0%), risk exposure plummets to under 1% once users cross a threshold of 10+ successfully completed loans. By providing a scaffolded environment to safely navigate this high-risk onboarding phase, Zen-i offers a scalable, practical framework to eliminate credit invisibility, reduce systemic consumer debt, and foster long-term financial mobility

1. Introduction

In the modern financial world, credit is as essential to financial mobility as oxygen is to human survival. It allows an individual to secure personal loans, obtain mobile contracts, or finance major life assets like a vehicle or a home. Credit bureaus and banking institutions rely entirely on these credit histories to assess an individual's financial reliability.

Traditionally, these scores are built on a rigid, predictable flow of monthly cash through conventional banking accounts, tracking how consistently outstanding balances are repaid. For a significant portion of the population, this foundational footprint is how long-term wealth and financial stability are unlocked.

In South Africa, this traditional framework leaves a massive portion of the population behind. Many citizens struggle to improve their credit scores or even establish a credit history due to existing debt or because they rely on flexible or non-traditional incomes.

The Scale of the Problem: A staggering 16 million South Africans are currently “credit invisible” (TransUnion, New Telco-Powered Credit Score Set to Transform Access to Finance for Millions of South Africans, 2025). Their micro-transactions, cash flows, or open accounts that do not fit into traditional credit-scoring models results in their financial reliability becoming completely unrecognized.

The biggest flaw in the current market is how existing Buy Now, Pay Later (BNPL) companies and micro-lenders handle late payments. Instead of helping users get back on track, these companies rely heavily on hitting people with massive, punishing late fees as a way to make extra money or protect themselves.

This creates a dangerous trap for the user: the moment a payment is even a few days late, the heavy penalty itself makes the overall debt too expensive to pay off. The system immediately pushes the user into deeper financial trouble, ruining their chances of building a good credit score.

This project addresses this critical gap by designing a specialized Buy Now, Pay Later (BNPL) platform engineered specifically for the South African economic landscape. The system acts as a dual-purpose financial bridge and credit-builder by focusing on four core objectives:

Algorithmic Communication: Serves as a fluid transactional bridge between the user and the merchant, facilitating seamless installment payments.

The Fintech Safety Net: Intervenes when a user faces a late installment, utilizing a third-party fintech mechanism to clear the merchant's balance and transition the user to a fair, interest-bearing structure rather than a crushing flat penalty.

A Credit Score Simulator: Provides a predictive internal scoring system that mimics real credit bureau behavior, giving users clear visibility into how their transaction habits translate to formal creditworthiness.

Systemic Debt Reduction: Actively combats South Africa's debt epidemic and erodes the 16-million-strong credit invisible population by funneling non-traditional cash-flow data directly into recognized credit scoring frameworks (TransUnion, TransUnion's 'Be the Reason Things Change' campaign advances financial inclusion in Africa, 2025)

Our main contribution:

- **Credit for Non-Traditional Income Earners:** We demonstrate how alternative cashflow generated through a BNPL system can be leveraged to establish formal credit histories for individuals with flexible incomes who are not qualified by traditional banking models

2. Related Work

Sezzle Up. A program developed by Charlie Youakim (Dean, 2026) that use BNPL to record the monthly installments paid by fixed-income users which is sent to credit bureaus for credit scoring.

MoreTyme: Offered by TymeBank, this tool splits your purchase into 3 payments over 60 days. It can be used both online and at physical retail stores. (MoreTyme, 2026)

Perpay (US): Perpay is unique because it functions as an e-commerce marketplace built entirely around building credit through BNPL.

Affirm (Global/US): Affirm is one of the most transparent lenders when it comes to credit reporting. Unlike apps that only report if you fail to pay, Affirm automatically reports both your positive (on-time) and negative payment history to Experian and TransUnion for most of its installment plans. (Monefy, 2026)

How our work differs. Existing tools only cater to specific income types and cater to different needs. We provide a safe space where inspiration of each of the tools listed above will be integrated into one application. Our target audience is catered to, but not only, individuals with flexible incomes to establish a credit score using cash flow information that is generated with a BNPL system.

3. Methods

3.1 System Architecture

The platform consists of two components:

Backend (Node.js / Express): We used Open Payments and Interledger Wallets to provide inclusivity and convenience for all types of income earners and payment services.

Frontend (React / TypeScript): We have two entities, a Merchant and Customer dashboard that caters to both users' needs.

3.2 Confidence Score and Buy Now Pay Later

Our system works where a confidence score mimics a realistic credit score based on how punctual users are to pay off their installments. A user will take out installments on products and services with the BNPL feature, and each installment will be paid off monthly. The system will record each paid off installment and calculate a realistic credit-based confidence score.

3.3 Open Payments

We have used Open Payment wallets (Interledger, 2020) as the target users of our system are those with flexible and non-traditional income types. It acts as a digital wallet to proxy the gap in financial inclusion.

4. Results

Establishing a credit footprint in South Africa is plagued by a classic catch-22: you need credit to build a credit score, but traditional lenders won't grant you credit without one. This systemic barrier is particularly devastating for first-time income earners, young adults, and gig economy workers. To justify locking these segments out, traditional financial institutions point to incredibly high initial default rates. However, this creates an aggressive bottleneck.

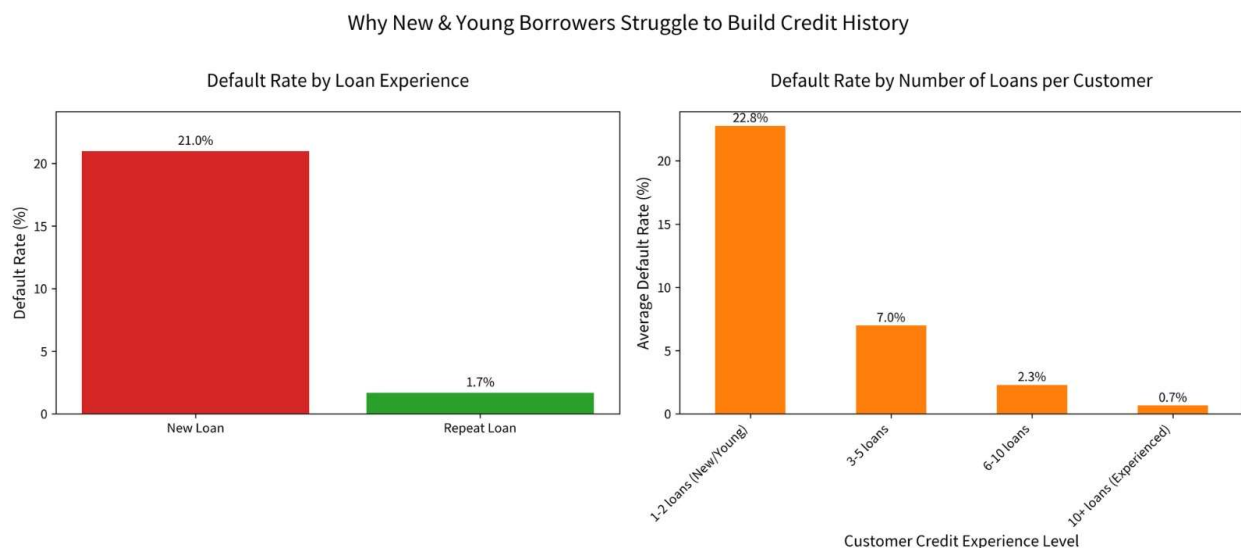


Figure 1: Analysis of default rates by loan experience and total number of loans per customer.

The graph above demonstrates the credit exclusion cycle that disproportionately affects new and young borrowers, including students, young adults, gig workers, and those with non-traditional incomes. New loans show a dramatically high 21.0% default rate — over 12

times higher than the 1.7% rate for repeat loans. Similarly, customers with only 1–2 loans (our proxy for inexperienced or young borrowers) face a 22.8% default rate, which drops sharply to under 1% as they build credit history with 10+ loans. This stark risk disparity leads traditional lenders to avoid this segment, preventing these individuals from accessing credit and building a positive credit history — effectively locking millions out of broader financial opportunities and perpetuating a cycle of financial exclusion.

5. Conclusion

This report presented Zen-i, an innovative Buy Now, Pay Later (BNPL) platform designed to dismantle the barriers of credit invisibility that currently marginalize 16 million South Africans. By leveraging an agile architecture built on Node.js, React, Open Payments, and Interledger Wallets, Zen-i establishes a fluid, inclusive transactional bridge optimized for both fixed and non-traditional income earners.

The broader implications of Zen-i lie in its power to disrupt this cycle of exclusion without plunging consumers into a debt trap. By replacing predatory, flat late fees with an interest-bearing "Fintech Safety Net" and utilizing a predictive Credit Score Simulator, the platform guides users safely through their high-risk borrowing phases. Instead of penalizing temporary cash flow fluctuations, Zen-i captures alternative data to turn everyday micro-transactions into verifiable creditworthiness. Ultimately, Zen-i provides a scalable, empathetic, and legally conscious framework capable of lowering systemic consumer debt, expanding financial inclusion, and fostering long-term economic mobility across South Africa.

Code and Data

- **Code repository:** <https://github.com/Plum165/Fintech-Hackathon-2026>
- **Data/Datasets:** <https://www.kaggle.com/datasets/moustapha96/african-credit-scoring/data?select=Test.csv>

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